

TERM PROJECT – PROJECT FINAL REPORT

1. GOAL

To present the term project final report, describing in detail the neural network-based (NN-based) approaches proposed, both **shallow** NN-based and **deep** NN-based approaches, to solve the stated problem, as well as the final results and conclusions.

2. LEARNING OUTCOMES

- To identify problems that can be solved using neural networks concepts and methods.
- To devise neural network-based solutions to problems, considering ethical, efficiency and quality criteria.
- To advance in executing your class projects in a planned and organized way to achieve the fulfillment of your goals
- To express your ideas in a clear and organized way, demonstrating argumentative capacity and communication skills in oral and written discourse.

3. DESCRIPTION

The goal of this assignment is to present the term project final report. In this report, you will need to describe in detail the neural network-based (NN-based) approaches proposed, both **shallow** NN-based and **deep** NN-based approaches, to solve the stated problem, as well as the final results and conclusions.

4. DELIVERABLES

A. *WRITTEN REPORT.*

A written report, which must be presented in the format “*Template to present project reports*” available in the section “Información general” of the Moodle course.

This document will need to include at least the following sections

:

PROJECT GOAL AND SCOPE. The goal of this project consists of a few sentences with an executive statement of the goal of the project.

LITERATURE REVIEW

This will help you answer the question about where to start your background research. It will need to include some basic definitions. Also, should mention key technical definitions to situate the project, as well as key preliminary facts about this problem. All this information with additional resources is a first step necessary to help you achieve your goals successfully.

CASE STUDY

To identify a case study, you should try to answer the question: Where can I find a partial solution/key advance on my way to the goal? A specific citation is required, where the case study is reported/documented. In addition, you need to specify the reasons why it would help your project.

TRAINING DATA SET

In most cases, you will need to use a data set to solve your problem. Thus, a general description of the data set to be used to train the neural network(s) must be provided. Otherwise, you will need to provide a general description of the approach used to solve the problem.

DATA PRE-PROCESSING AND EXPLORATORY DATA ANALYSIS: You will need to perform data cleansing on your data set and perform some basic statistical analysis to summarize the main characteristics of your dataset, you are expected to illustrate your analysis using visual methods. Namely, figure out which features are continuous, which ones are categorical, check if there exist some evident relationships between the features, take a look at the distribution of each feature, etc. Also, remove useless features and values, and find better representations of the features to feed your model, scaling the features, and so on.

For some basic information on exploratory data analysis, please see the material posted in the section “Introducción al Aprendizaje Automático” of the Moodle course.

PROPOSED NEURAL NETWORK BASED APPROACHES: In this final report you will need to present the application of **both a shallow NN-based and a deep NN-based approach**. Precisely describe the type of neural networks and particular architectures used, explaining the reasons for your choice. For the shallow neural networks, you will need to explain the feature engineering process, particularly, if you applied feature selection or feature extraction, and specifying the features that are given as input to the shallow neural network. Also, for both approaches, the network learning and testing process need to be explained, as well as the performance measures used to evaluate the performance of the implemented neural networks.

PERFORMANCE EVALUATION AND FINAL RESULTS

Describe the results obtained. Accurately estimate how well your proposed method is doing by applying suitable performance measures. Determine whether your method overfits or underfits; you should be able to diagnose if your model is over- or underfitting the data and take actions to fix the problems with your model.

ETHICAL CONSIDERATIONS/QUESTIONS

You will need to ask yourself questions regarding the process of developing your project and the impact of your solutions. In general, ethical considerations in research are a set of principles that guide your research designs and practices. These principles include voluntary participation, informed consent, anonymity, confidentiality, potential for harm, and communication of the results.

WEB LINKS TO SOURCE CODE AND EXPLANATORY VIDEO

- **SOURCE CODE**

You will need to provide the access to the source code (Colab notebooks or others) and other documents of your project. Here, you will need to provide the web access (URL) to where you have organized your implementations of the solution to your problem. To handle the data, software and documentation of your term project, you are advised to use a Github repository, though, alternative repositories can be used. The information needs

to be organized in an intuitive/structured manner. Again, the information to access such repository must be included here.

- **EXPLANATORY VIDEO**

In this section, you will need to include the information to access the explanatory video of your project final report (on YouTube or where applicable). Please see some instructions to prepare such video below.

CONTRIBUTIONS OF THE TEAM MEMBERS

The roles, activities and contributions of each of the team members during the development of this delivery must be concisely described. Please keep in mind that in the different deliveries, you could rotate the roles so that everyone takes a turn and thus gains experience of the different roles. Also, you may want to allocate the crucial roles within the team to the people who would like to take them, are best qualified to carry them out through prior or current experience, or even to people who would like to take on an unfamiliar role in order to gain experience of performing that role.

However, each team will need to select a **team coordinator**, who besides supervising and motivating the members of the working group will be the direct communication channel between the professor and the team.

The teams are expected to internally solve any issue or conflict that arises among the members of the team. However, if an issue or conflict could not be solved inside the team, the team coordinator is responsible for reporting it to the professor.

The contributions of each team member may be provided in a table as shown below.

Team member	Role	Activities/contributions	

More information about such roles can be found in the document “sobre roles y actividades de miembros equipos de trabajo – proyecto.docx” published in the same section in the Moodle course.

CONCLUSIONS AND FUTURE WORK

Devise the conclusions of your work, address the limitations of the proposed approach and how it could be enhanced or built on it.

USE OF ARTIFICIAL INTELLIGENCE TOOLS (PREPARING THIS DELIVERABLE)

As a reminder, you are allowed to use Artificial Intelligence (AI) tools, such as ChatGPT, to help gather information and generate ideas or source code, but you may not copy and paste information directly from the AI tool and present it as your own without citing it. You are responsible for the information you submit based on an AI query and for ensuring that it does not contain misinformation, unethical content, or violate intellectual property laws.

You must properly document and cite the use of artificial intelligence tools to ensure academic integrity. When applicable, you are expected to include a statement in your assignment describing which AI tool you used and how you used it. For example, *"ChatGPT was used to draft*

approximately 50 percent of this document and to provide review assistance. AI-produced content was edited for accuracy and style".

REFERENCES

Include the most relevant sources that you used in the development of your project. Each source you cite in the paper must appear in your reference list; likewise, each entry in the reference list must be cited in your document.

A. ACCESS TO SOURCE CODE IMPLEMENTATION

You will need to provide the source code (Colab notebooks or others) with the implementation of your **shallow and deep** neural network based approaches. To handle the data, software and documentation of your term project, you are advised to use a Github repository, though, alternative repositories can be used. The information needs to be organized in an intuitive/structured manner. The information to access such repository must be included in the written report as stated above.

B. EXPLANATORY VIDEO OF YOUR REPORT

You will need to present the final project progress report in a video that does not exceed 6 minutes. Thus, we suggest that you prepare a presentation of no more than 12 slides, which will emphasize:

- i) the proposed deep neural network approach,
- ii) performance measures,
- iii) a comparison between the shallow and deep neural network approaches,
- iv) final results, and
- v) conclusions and future work.

You can use this presentation as a basis to prepare your video. The URL to access this video must be informed on the corresponding section of the written report as explained above.

To help you prepare your video, you are advised to see the tutorial: **Effective Presentation Skills Tutorial** available at <https://www.niu.edu/presentations/index.shtml>

Please be aware that though some of the tips in this tutorial are intended to in-person presentations, most of them still apply for on-line presentations.

Very important: Your team will need to participate in the oral discussion of your term project final report in class.

5. SUBMISSION

The file of the term project final report, in pdf format, needs to be submitted via Moodle. Only **one submission per team** is required. Make sure ONLY ONE COPY of the deliverable is submitted, thus, you will need to coordinate within your team who will be in charge of submitting the assignment. Students are advised to retain a copy of every assignment submitted for their own records.

- **File to submit:**

Written report: a document with the details of the term project final report.

Please make sure that in this report you include the web links to:

- ✓ the repository where your source code is available, and
- ✓ the explanatory video, a video presenting your term project accomplishments.

- **File naming:**

The team must use the following file naming convention for the assignment:

Project proposal: “TP-final-” appending your team ID, where team ID is the number assigned to your team. For example, “TP-final-5.pdf” for team ID 5.

6. DEADLINE (DATE/TIME)

Written report:

9:00pm, Tuesday, May 26th, 2026

Oral presentation:

Presentation video: 9:00pm, Tuesday, May 26th, 2026;

Oral presentation/discussion in class: Wednesday, May 27th, 2026.